

**INFANT MORTALITY CHANGES FOLLOWING THE
THREE MILE ISLAND ACCIDENT**

**Dr. E.J. Sternglass
Department of Radiology
University of Pittsburgh
School of Medicine
Pittsburgh, PA 15261**

**Presented at the 5th World
Congress of Engineers and
Architects
Tel-Aviv, Israel
January 25, 1980**

Summary

An examination of the monthly changes in infant mortality in Pennsylvania and the nearby areas of Upstate New York as given in the U.S. Monthly Vital Statistics reports indicate that the mortality rate rose significantly shortly after the Three Mile Island accident in the directions where the plume of radioactive gases was known to have moved. The number of reported infant deaths per month rose from a minimum of 141 in March of 1979 just before the accident to a peak of 271 in July, declining again to 119 by August. This is an unprecedented and highly significant rise of 92% in the summer months when infant mortality normally reaches its lowest values. In the four month period following the accident, there were 242 deaths above the normally expected number in Pennsylvania and a total of about 430 in the entire Northeastern area of the United States. The hypothesis that this abnormal rise was associated with the gaseous releases from Three Mile Island is shown to be strongly supported by the following considerations. First, large amounts of radioactive Iodine-131 were released from the plant, estimated by the utility's own radiological consultants to have amounted to 14 curies, together with 10 million curies of other fission gases, most of the activity escaping in the first two days before the order to evacuate pregnant women and young children was issued. Secondly, infant mortality peaked three to four months after the initial releases took place. This corresponds to the period required for infants to be born whose thyroid glands were most active in taking up the radioactive iodine while producing growth hormone when the accident occurred, thus explaining the large rise in the number of immature and underweight infants that died of respiratory distress as indicated by an examination of hospital records. Thirdly, the greatest rises took place in areas closest to the plant, decreasing with distance away from Harrisburg and the state of Pennsylvania, until for states well to the west and south, there was a decline in infant mortality rates. Thus, while Pennsylvania

increased from 10.4 per thousand live births in March to 18.5 in July of 1979, the United States rate as a whole declined from 14.1 to 12.5. The rise moved Pennsylvania from well below the U.S. average to the highest infant mortality rate for any state east of the Mississippi River. Other evidence supporting this conclusion is discussed, including the occurrence of abnormal increases of infant mortality rate in areas that received heavy fallout from nuclear weapons tests during the 1950's and 60's when Iodine-131 levels reached comparable levels, as well as similar effects in areas close to other nuclear reactors known to have released comparable amounts of radioactive gases from damaged fuel elements over a period of years in the course of normal operations. The implications for further health effects due to cancer and other diseases are discussed.

Although the major emphasis in estimates of the health impact produced by the accident at Three Mile Island has been on the increase in cancer risk and the effect of the psychological stress⁽¹⁾, the greatest immediate concern was connected with the possible effects on the developing infant in the mother's womb due to the known tendency of radioactive iodine to concentrate in the fetal and infant thyroid glands⁽²⁾.

It was the existence of uncontrolled large releases of fission-produced radioactive gases including Iodine-131 that led Governor Thornburg of Pennsylvania to order the evacuation of all pregnant women and children below school age from within a radius of five miles around the plant on the third day of the accident, Friday, March 30, 1979.

Subsequent studies by various government and private organizations have confirmed that large quantities of radioactive Iodine-131 were in fact released in the course of the accident, the estimates ranging from 1.4 curies in an early study prepared by the NRC⁽³⁾ to as high as 14 curies of Iodine-131 in a later study by a private consulting firm for the utility⁽⁴⁾. To appreciate the significance of these amounts, it is only necessary to point out that the unit which is used to measure concentrations of Iodine-131 in milk is the pico-curie, which is one millionth of one millionth of a curie.

These studies further showed that most of the gaseous releases took place in the first two days after the accident that began in the early morning of March 28. Thus, it was calculated that of the 10 million curies of radioactive gases released in the first five and a half days, 7 million curies has been emitted in the first 36 hours.⁽⁴⁾

Likewise, most of the external gamma ray dose due to the passing clouds of fission gases estimated by all organizations to have been in the range of 70 to 80 millirems to individuals nearest the plant was found to have been received in the first few days of the accident.

These studies further showed that most of the thyroid dose was received through the inhalation of radioactive iodine isotopes rather than through the ingestion of milk or water⁽⁴⁾.

For the infant thyroid, the maximum dose was calculated to have been of the order of 10 mrem by inhalation and 1.1 mrem by ingestion of milk⁽⁴⁾. Due to the smaller volume of air inhaled by infants, the maximum dose was found not to have been much larger than for adults despite the ten-fold smaller size of the infant thyroid.

No estimates were published for the fetal thyroid, which is known to begin functioning actively in about the fifth to sixth month of intra-uterine development.⁽⁵⁾⁽⁶⁾ However, earlier measurements carried out for comparably small exposures during periods of heavy fallout from nuclear weapons testing clearly indicate that fetal thyroid doses can be some 10 to 20 times larger than for infants or 100 times that of adults⁽⁷⁾⁽⁸⁾. This arises from the fact that the early thyroid gland has a very small mass of the order of 0.10 to 0.20 grams, or some one to two hundred times smaller than the 20 gram adult human thyroid.

Thus, it is possible to estimate that the thyroid gland of fetuses in the more heavily exposed areas within five to ten miles of the Three Mile Island plant are likely to have received radiation doses of the order of 100 to 1000 mrem from Iodine-131 alone. To this must be added the whole-body gamma dose from noble gases and the contributions from the many short-lived iodine isotopes as well as the whole body doses from the other important isotopes such as Cesium-137, Cesium-134,

Tritium, Barium-140, Strontium-89, etc., depending upon the mother's diet in the weeks following the accident.

Thus, typical thyroid doses 200 to 1100 (in excess of 200) mrems are likely to have been received by developing infants in their mother's womb for those in their 5th to 9th months of development. Due to the fact that the radioactive gas plume frequently touched the ground within a few miles of the plant, individuals in these areas could easily have received five to ten times higher doses than the average in the first one or two days of the accident, before the evacuation of pregnant women was ordered.

A dose of the order of 200 to 1100 mrems is comparable with that received by the fetus in the course of diagnostic x-ray examinations since a single abdominal film gives a fetal dose of the order of 200-300 mrem⁽⁹⁾ ⁽¹⁰⁾.

The fact that diagnostic x-ray exposures during fetal development can lead to serious biological damage is by now widely recognized in the medical community as a result of the large scale epidemiological studies of Stewart⁽¹¹⁾, MacMahon⁽¹²⁾, Diamond⁽¹³⁾, Bross⁽¹⁴⁾, Graham⁽¹⁵⁾ and others.

Not only is there an increased risk of leukemia and cancer as first discovered by Stewart⁽¹¹⁾ but the more recent prospective study of Diamond and his associates at Johns Hopkins University⁽¹³⁾ revealed an increased risk of death from all causes for those who received x-rays in utero, primarily within the first year of life. These included deaths from respiratory system problems, infectious diseases and diseases of the central nervous system. Furthermore, both the Johns Hopkins study and the study of Stewart⁽¹¹⁾ revealed that the earlier the exposure takes place, the greater is the risk of adverse effects.

Thus, Stewart's study showed that whereas the dose needed to double the risk of leukemia and cancer was of the order of 1,200 mrem just before birth when most x-rays were taken, a mere 80 mrem doubled the risk when x-rays were given in the first three months of pregnancy⁽¹¹⁾.

Thus, doses of 200 to 1100 mrem to the critical thyroid gland controlling the growth and development of the fetus beginning in the 5th to 6th month of pregnancy could be expected to produce a significant effect on the fetus through retardation in growth and maturation, thereby increasing the risk of mortality within the first year of life. The greatest mortality occurs during the critical period of transition from intra-uterine existence when the lung of the newborn must suddenly take over the life-sustaining function of respiration⁽¹⁶⁾.

A large number of studies have shown that infants that are born immature, underdeveloped or underweight have a much higher incidence of respiratory distress or hyaline membrane disease⁽¹⁶⁾. Thus, even a small degree of retardation in development due to a reduced output of growth hormone by the thyroid gland during the last three to four months of intra-uterine development would be expected to increase the risk of death due to respiratory insufficiency immediately after birth. Failure of the critical lung surfactant to be produced in adequate amounts can therefore lead to respiratory problems and death as a result of damage either to the thyroid or the pituitary gland which in turn controls the thyroid's output of growth hormone⁽¹⁷⁾.

The fact that radioactive iodine from nuclear weapons tests can indeed reduce human fetal and infant growth has been observed as a result of the accidental fallout exposure of the Marshallese Islanders following the large hydrogen bomb test BRAVO in 1954⁽¹⁸⁾. Not only was there an increase in fetal deaths among the women exposed at that time, but there was also a severe reduction in growth of the children exposed to fallout, the effect being greater the younger the children were.

The effect of even small doses of radioactive iodine on fetal development is further supported by the fact that the birth-weight of babies born in the United States suddenly began to decline during the 1950's, ⁽¹⁹⁾ the period when the heaviest

fallout from weapons testing occurred in the continental United States. Furthermore, since the end of large-scale atmospheric bomb-tests by the U.S. and the U.S.S.R., there has not only been an end to the trend towards smaller birth-weight but there was also an unexpected resumption in the decline of infant mortality which had also halted during the period of heavy fallout⁽²⁰⁾ ⁽²¹⁾.

This is shown graphically in Figure 1 taken from the most recent summary of infant mortality trends published by the U.S. Department of Health, Education and Welfare's National Center of Health Statistics⁽²²⁾. As shown by the arrows added to indicate periods of heavy nuclear testing in the atmosphere, the decline of infant mortality rates in the United States came to a sudden halt between 1945 and 1951, when the testing of nuclear weapons at Alamogordo, New Mexico, in the Marshall Islands, and in Siberia began.

Between the years of heaviest weapons testing in the lower atmosphere namely 1956 to 1958, there was actually a reversal of the previous downward trend of infant mortality, which ended only when nuclear testing was temporarily halted during the years 1959 to 1961, after which there was a brief renewal of bomb testing that did not end until the 1963 test-ban treaty was concluded.

Although France and China continued to test nuclear weapons in the atmosphere, the end of large atmospheric tests by the U.S., U.S.S.R. and U.K. led to a gradual reduction in infant mortality rates, until in the last few years, both white and non-white populations showed a resumption of the pre-testing rate of decline that had so dramatically halted during the 1950's for reasons that were not understood at the time⁽²³⁾.

A connection between fallout and infant mortality changes was first suggested in 1969⁽²⁰⁾. However, at that time the evidence that the fetus in the early phases of development may be fifteen times more sensitive to radiation than the full term infant had not yet been published by Stewart and Kneale⁽¹¹⁾ so that it was difficult to explain the large magnitude of the effect⁽²⁴⁾. Also, the study of Diamond and his coworkers had not yet been published showing that all causes of death and not merely leukemia and cancer were increased by small intra-uterine exposures, so that a generalized effect on all forms of mortality was unsubstantiated.

As a result it was argued by some authors that the halt in the decline of infant mortality might be due to the fact that all possible effects of improved medical care, diet and drugs had been exhausted, and that therefore infant mortality had gone down as far as it would ever be able to decline in the United States⁽²⁴⁾.

However, as Figure 1 clearly shows this did not turn out to be the case: infant mortality started to decline once again after heavy fallout ended and in the most recent years has actually accelerated its rate of decline, rapidly approaching the projected rate that would have existed if there had been no halt during the time of massive nuclear weapons tests⁽²¹⁾.

The effect of small amounts of short-lived fission products such as I-131 as distinct from the long-term trend that is correlated with the build-up of Strontium-90 in the bone of young women prior to pregnancy⁽²¹⁾ is shown particularly strikingly in Figure 2.

Here, the infant mortality rate for New Hampshire some 2,500 miles to the east of the Nevada test site has been plotted for the years 1946 to 1974, together with the officially announced yields of small tactical nuclear weapons detonated in Nevada between 1951 and 1962⁽²⁵⁾.

Inspection of this plot shows that simultaneously with the onset of Nevada tests in 1951, there was a sudden halt in the long-term decline of infant mortality followed by brief periods of small reversals in the downward trend which occurred in the same years as the nuclear weapons tests in Nevada. Whenever there was a test in Nevada, there was a sharp rise in infant mortality in New Hampshire, where the fallout came down with the heavy rain and snow falls in the White Mountains.

Not until the Nevada tests ended did these highly abnormal spikes in infant mortality disappear, and only after the end of all large scale nuclear weapons tests did the infant death rate decline once more, finally reaching the low values that would have been reached had the pre-bomb testing rate of decline continued.

The significance of these results for the case of the Three Mile Island accident is that the external gamma ray doses from bomb fallout in the eastern United States in any given year were generally smaller than external gamma doses of 70 to 80 mrem measured in a few days at Three Mile Island.

Thus, the long-term gamma dose measurements carried out at the Brookhaven National Laboratories since the late 1940's⁽²⁶⁾ show that only in 1963 did the annual gamma ray dose reach 76 mrem, the same dose experienced at Three Mile Island in a matter of days.

Yet, both Figure 1 and 2 clearly show significant peaks in infant mortality associated with weapons tests in Nevada, the Pacific and Siberia. This indicates that the actual internal doses to critical organs from inhaled and ingested fission products far outweigh in their biological importance the external gamma dose recorded by badges and survey meters whenever fission products are released into the atmosphere⁽²⁷⁾.

This is supported by a more recent study that found a correlation of changes in infant mortality in different sections of Wisconsin with measured changes in Sr-90 levels in the milk following bomb-tests and the start-up of nuclear power plants⁽²⁸⁾.

Clearly, whatever the still not fully understood biological mechanisms are that cause prematurity and early neonatal death associated with respiratory distress, external gamma doses from fission products containing I-131 of the order of 10 to 50 millirems were associated with yearly infant mortality changes of the order of 5 to 25% in the eastern U.S. during the late 1950's, that is changes of about 0.5%/mr

It follows that the releases of fission products containing I-131 in the Three Mile Island accident should also be followed by detectable peaks in infant mortality associated with immaturity and respiratory distress within a few months after the accident, when the infants whose thyroids had begun to function when the iodine was released were born. Furthermore, since fetuses of less than 5-6 months gestational age do not have fully developed thyroid gland able to concentrate I-131 to the same extent as those that are present in the last 3 months of gestation, (5) there should be a rapid decline in newborn deaths beginning in the third and fourth month after the end of the gaseous releases.

That this is indeed the pattern of infant mortality changes that has taken place following the Three Mile Island accident is apparent from the following considerations.

Turning first to the data on the month-by-month infant mortality in Pennsylvania as reported in the Monthly Vital Statistics reports published by the National Center for Health Statistics (29) plotted in Figure 3, one sees that a large peak in the infant mortality rate occurred in the summer of 1979, some three to four months after the releases took place. Grouped in 3 months periods, the rise was 32% (Table I).

Compared with the normal pattern of infant mortality throughout the year, which generally declines in the summer months when the risk of pneumonia and influenza is small, the 78% rise from a rate of 10.4 per thousand live births in March to a

peak of 18.5 in July, followed by a decline to the normally expected low of 8.5 in August, is totally anomalous. However, it is completely consistent with the expected action of radioactive Iodine-131 on the fetal thyroid.

The rise from 141 to 271 infant deaths is statistically highly significant representing an increase by more than 5 standard deviations for which the probability of a purely statistical fluctuation is much less than 1 in 1000 ($P \ll 0.001$).

Such a large rise did not occur for the United States as a whole as shown in the plot of Fig. 4⁽²⁹⁾. Furthermore, it is seen that there was no such large rise in mid-summer in previous years as illustrated for 1977 and 1978 in Figure 4.

In fact, the July infant mortality rate of 18.5 per 1000 live births in Pennsylvania was the highest for any state in the entire United States east of the Mississippi, moving Pennsylvania above such rural states as Mississippi and Alabama, where the large non-white population with its poorer socio-economic status, medical care and diet had traditionally been associated with a much larger infant mortality⁽³⁰⁾. (Table II).

Whereas prior to the Three Mile Island accident, Pennsylvania had an infant mortality rate well below the U.S. average, by July it had moved well above the U.S. average, returning to its normal position below the U.S. mean in August, 4 to 5 months after the accident. (See Figure 5). This was the time when infants were being born that did not have a fully developed active thyroid gland at the time of exposure.

Since for the first few days when the initial release occurred, the winds were mainly towards the West, North-West and North⁽³⁾, one would expect significant increase in infant mortality in the summer months in upstate New York and Western Pennsylvania, but not in New York City directly to the East. (See maps of Figure 6 and 7). At the same time, the Harrisburg area should show the greatest increase in the newborn mortality rate.

This expectation is born out by the changes in infant mortality for three month periods before and after the Three Mile Island releases ended, (Fig. 8 and Table III). In this table, the data from the U.S. Monthly Vital Statistics is listed together with data obtained for a major hospital in each of Harrisburg and Pittsburgh, ⁽³¹⁾ since the county-by-county data in Pennsylvania were not yet available.

As can be seen from Table IV and Figures 8 and 9, there was a 7 fold rise or a 630% increase in the number of newborn deaths associated with immaturity and respiratory diseases in the Harrisburg hospital for the three month period of May-June-July 1979 relative to both the same period in 1978 and the immediately preceding three month period of February-March-April.

A similar pattern is seen to have occurred in the Pittsburgh area as reflected in the data for a major hospital that accounts for nearly half of all deliveries (Table V. and Fig. 10) though with a smaller percent change. Again, there is an unusual summer peak in 1979 that did not occur in the previous year. In fact, examining the case for the peak month of July in Figure 11, it is seen that both the total number of deaths and the rate per thousand births was declining before 1978, rising again from 8 deaths in July of 1978 to 24 in July of 1979.

This unusual peak of infant deaths in July was also observed in Northern New England but not in Southern New England, as expected from the known northward direction of the winds in the first few days of the accident. (See Table VI).

Although the increases in the number of deaths among newborns in the Harrisburg Hospital are small in absolute numbers, they are nevertheless highly significant statistically. The average number of newborn deaths per month during 1978 was 0.42 per month with a standard deviation of ± 0.2 , while the average for the summer months was only 0.33 per month. Thus four such deaths in June of 1979, exactly three months after the accident, represents a 10 to 12 fold increase above

normal expectations, whose occurrence by mere chance is much less than 1 in 100.

Combined with the geographical pattern of infant mortality changes from Ohio to New England, and the absence of any similarly large summer peak of infant mortality in the previous years, it seems inescapable to conclude that the accident at Three Mile Island did in fact lead to a significant increase in infant mortality similar to those encountered from earlier comparable episodes involving fission products from nuclear weapons testing and releases from nuclear plants.

For the 3 month period May through July 1979, the total number of excess deaths was 352 compared with that normally expected in the states of Pennsylvania, Ohio and New York, using the rate of decline for the United States as a whole^(†).

Comparable percent increases are likely to occur for childhood cancer and leukemia in the next ten to fifteen years^{(32) (33)}, as well as somewhat lower percent increases in all causes of death in the much larger general population according to the studies of Lave and his associates for the case of 50 metropolitan areas in the United States during the heavy fallout period of the early 1960's⁽³⁴⁾. It would therefore appear that the Three Mile Island accident will turn out to have produced the largest death-toll ever resulting from an industrial accident, with total deaths from all causes likely to reach many thousands over the next 10 to 20 years.

Since for every infant that dies as a result of retarded growth and development, there are many more individuals (perhaps ten times as many) that are minimally damaged in their physical and mental abilities^{(16) (35)}, the effect on society will be much larger than premature death alone. In the light of these findings, it must be of great concern that the releases of radioactive gases and iodine from routine operations of nuclear reactors such as the Millstone Plant in Connecticut have been comparable in magnitude with

[†] If one adds the excess deaths in April, there were 40 in Pennsylvania and 35 in Upstate New York for a total of 427.

the releases from Three Mile Island, namely 3 million curies of mixed fission gases and 10 curies of I-131 in 1975 alone⁽³⁶⁾.

The full magnitude of the danger to the biological survival of nations that choose to pursue the development of nuclear reactors near their large population centers is therefore only now beginning to become apparent.

This is especially brought out by the fact that the greatest number of deaths actually occur in areas ten to one-hundred miles distant from the reactor site so that even a decision to locate nuclear reactors only in less populated areas would not appreciably change the mortality figures either for the effect of normal releases or major accidents. Nor would an evacuation plan for the population within 5 or 10 miles of the reactor significantly affect the overall number of deaths resulting from a repetition of the Three Mile Accident, since most of the deaths occur in the more distant areas due to the larger total populations they contain⁽³⁷⁾, though the greatest individual risk exists nearby.

This is illustrated by the observed infant mortality changes in the State of Rhode Island located between 20 and 50 miles downwind to the east of the Millstone Reactor near New London, Connecticut as shown in Figure 12.

In this plot, the infant mortality rate for Rhode Island has been plotted for the years 1965 to 1976, together with the releases of radioactive fission gases as published by the Nuclear Regulatory Commission (NRC)⁽³⁶⁾ since the start of operations in 1970.

For comparison, the infant mortality rate for the State of New Hampshire more than 150 miles to the northeast is also shown in this plot.

Inspection of Figure 12 shows that in the years prior to the start of the Millstone Plant in 1970 and the Connecticut Yankee plant at Haddam Neck in 1968, both states showed the same infant mortality rate.

However, beginning in 1970, the rate of decline of the Rhode Island infant mortality slowed down, while that of New Hampshire kept declining rapidly, leading to the development of a large gap in the mortality rates between the two states by early 1976. In the first quarter of that year, Rhode Island had a rate of 18.6 practically equal to its 1967 level of 19.3, while New Hampshire declined to an infant mortality only half as great, or 9.3 per 1000 live births.

The fact that the emissions from the Millstone plant are likely to be responsible for this dramatic gap is illustrated by Figure 13 which is a plot of the percent excess infant mortality in Rhode Island relative to New Hampshire versus the announced gaseous releases from the Millstone plant. A two-year moving average of the emissions is used in this plot to take account of both the short-lived I-131 and the longer-lived Cs-137 and Sr-90 in the releases that build up in the environment.

It is seen that the data fit a logarithmic or fractional power law of the form aD^x with $a = 38\%$ and $x = \frac{1}{2}$. This is the type of law expected on the basis of a free-radical type of indirect chemical action on cell membranes discovered by Petkau⁽³⁹⁾, which rises more rapidly at low doses and low dose-rates than at high doses and dose-rates. It is also the type of dose-response relation expected on the basis of a wide difference in sensitivity of different individuals in the population as discussed by Baum⁽⁴⁰⁾, Bross⁽⁴¹⁾ and Morgan⁽⁴²⁾, or a type of dose-response that leads of a large underestimate of health effects at low doses from data taken at high doses⁽⁴³⁾.

Based on an external gamma dose of 500 mr for the maximum permissible release of some 25 million curies per year⁽³⁸⁾, the slope of the dose-response curve increases from 2%/mr at 1 million curies per year to about 6%/mr at 0.1 million curies per year.

Based on the external dose from the gas-cloud alone, one would therefore expect an increase in infant mortality for the case of the Three Mile Island accident of 38×10^4 or 120% in an area some 20-30 miles distant. This is in good agreement with the three month increase in newborn infant mortality in Harrisburg of 600%, which reduces to 150% when averaged over the period of a year.

Thus, both qualitatively and quantitatively, the releases from normal operation from the Millstone Plant and the accompanying increases in infant mortality in Rhode Island some 20 to 30 miles downwind correctly predict the infant mortality changes in the Harrisburg area from the comparable noble gas and iodine discharges at Three Mile Island.

The importance of releases of Cesium-137 and Strontium-90 in addition to Iodine-131 for the more distant large metropolitan areas is illustrated by the plots of the Cs-137 and Sr-90 spatial distribution around the Millstone plant as taken from the utility's own environmental reports to the NRC⁽⁴⁴⁾ shown in Figures 14 and 15. The 4 to 5 fold greater concentration of both of these isotopes in the milk near the stack of the plant as compared with the more distant areas strongly argues against the interpretation advanced by the utility⁽⁴⁵⁾, the NRC⁽⁴⁶⁾ and the EPA⁽⁴⁷⁾ that these levels can be explained by fallout from nuclear weapons tests.

The hypothesis that these highly localized levels are indeed due to plant emissions is also strongly supported by the fact that the changes in concentrations of these isotopes in the milk throughout the year are greatest near the plant in the summer months following spring refueling, and least at more distant locations such as Delaware along the Atlantic Coast as seen in Figure 16 for Cs-137. Particularly significant is the fact that both within 10 miles of the plant and in Rhode Island, there was a peak in Cs-137 concentrations in the summer of 1976 before the heavy

Chinese fallout arrived in October. It should be noted that this fallout episode was also followed by a sharp rise in infant mortality along the north-east coast 3-4 months later.

The fact that the increases in cancer mortality around the Millstone Plant after it began operating have the same geographical distribution as the measured CS-137 and Strontium-90 concentrations in the milk, and the fact that the increase in infant mortality in Rhode Island is correlated with the releases from the Millstone Plant indicates that there is likely to be a significant cancer increase from the Three Mile Island accident.

But perhaps the most significant confirmation of the serious biological effect of routine radioactive gas releases from nuclear reactors is the plot of infant mortality rates at different distances from the Millstone Plant for July 1978 shown in Fig. 18, before the releases from Three Mile Island affected the mortality rates in New England.

It is seen that while New Hampshire, which has no operating nuclear reactor, declined to a low of only 1.9 infant deaths per thousand births from 21 in 1965, Rhode Island, directly downwind and close to both the Connecticut Yankee and Millstone Reactors, showed a ten-fold greater infant death rate of 19.9, completely nullifying 13 years of advances in public health and medical care.

Vermont, whose only nuclear reactor is located in its most southernly tip so that most of the state is not affected by the releases, reached a low of 4.2 per 1000 births. But all the other New England states with more nuclear generation of electricity, showed higher infant mortality rates than either New Hampshire or Vermont.

For these reasons, it appears doubtful whether any efforts to improve the safety of nuclear reactors, changes in their locations, or evacuation plants can significantly alter the public health impact of normal releases and such potentially more serious

accidents as occurred at Three Mile Island, where only a small fraction of the total fission products generated actually escaped into the environment. Needless to say, the mortality changes following Three Mile Island would be enormously multiplied in the event of a complete melt-down and rupture of the containment, threatening areas as large as the entire eastern United States with enormous damage in terms of cancer and thyroid damage for all age groups out to distances of hundreds of miles (37).

Since neither coal nor oil nor gas nor solar energy lead to the large-scale production of short-lived radioactive iodine and other fission products that seek out the critical organs of the developing infant, it would seem prudent to apply future technological efforts to the more forgiving sources of energy that do not threaten the most precious of our nation's resources: the physical and mental well-being of our children for generations to come.

References

1. "Report of the President's Commission on the Accident at Three Mile Island", John G. Kemeny, Dartmouth College, Chairman, 1979.
2. Chapman, E.M., Corner, G.W., Jr., Robinson, D., and Evans, R.D., "Collection of Radioactive Iodine by Human Fetal Thyroid", J. Clin. Endocrinology, Vol. 8, p. 717, September 1948.
3. "Preliminary Estimates of Radioactivity Releases from Three Mile Island", L.H. Barnett, Environmental Eval. Branch, Division of Operating Reactors, Office of Nuclear Reactor Reg., N.R.C., Washington, DC 20555, April 12, 1979.
4. "Assessment of Offsite Radiation Doses from the Three Mile Island Unit 2 Accident", Pickard, Lowe and Garrick Inc. for the Metropolitan Edison Company, July 31, 1979, TDR-TMI-116.
5. Dyer, N.C. and Brill, B., "Fetal Radiation Dose from Maternally Administered Fe-59 and I-131", Proc. 9th Ann. Hanford Radiobiology Symposium, Richland, Washington, March 5-8, 1969.
6. Dawes, G.W., "Fetal and Neonatal Physiology", Year Book Medical Publishers, Inc., Chicago, 1968.
7. Beierwaltes, W.H. et. al., "Radioactive Iodine Concentration in the Fetal Human Thyroid Gland from Fallout", J. Am. Med. Assoc., Vol. 173, p. 1895, August 27, 1960.
8. Eisenbud, M., "Radioactivity in the Environment-Radioiodine Concentration in the Fetal Thyroid", Pediatrics (Supplement), 41, Part II, p. 174, January 1968.
9. "Handbook of Selected Organ Doses for Projections Common in Diagnostic Radiology", M. Rosenstein, HEW Publication (FDA) 7-8031, May 1976, P.H.S., Bureau of Rad. Health, FDA.
10. Sternglass, E.J., "Cancer: Relation of Prenatal Radiation to the Development of the Disease in Childhood", Science, Vol. 140, p. 1102, 1963.
11. Stewart, A., and Kneale, G.S., "Radiation Dose Effects in Relation of Obstetric X-Rays and Childhood Cancers", Lancet, Vol. 1, pp. 1185-1188, 1970.
12. MacMahon, B., "Prenatal X-Ray Exposure and Childhood Cancers", Journal of the National Cancer Institute, Vol. 28, p. 1173, 1962.
13. Diamond, E.I., Schmerler, H., and Lillienfeld, A.M., "The Relationship of Intrauterine Radiation to Subsequent Mortality and Development of Leukemia in Children", American Journal of Epidemiology, Vol. 97, p. 283, 1973.
14. Bross, I.D.J. and Natarajan, N., "Leukemia from Low Level Radiation", New England Journal of Medicine, Vol. 287, p. 107, 1972.

15. Lyon, J.L., et. al., "Childhood Leukemia Associated with Fallout from Nuclear Testing", New England Journal of Medicine, Vol. 300, p. 397, 1979.
16. Cavanagh, D. and Talisman, M.R., "Prematurity and the Obstetrician", Appleton-Century-Crofts Div. of Meredith Corporation, New York, 1969.
17. Knelson, J.H., "Environmental Influences on Intrauterine Lung Development", Archives of Internal Medicine, Vol. 127, p. 421, March 1971.
18. Conard, R.A., Rall, J.E., and Shitow, W.W., "Thyroid Nodules as a Late Sequel of Radioactive Fallout", New England Journal of Medicine, Vol. 274, p. 1392, 1966; also, "Growth Status of Children Exposed to Fallout Radiation on the Marshall Islands", Pediatrics, Vol. 36, p. 721, 1965. A general slowing down of growth in the United States beginning for children born after 1945 has been observed by Damon, A., Am. J. Phys. Anthropology, Vol. 29, No. 1, 45 (1968) and in Social Biology, 121(1), 8, 1974.
19. Chase, Helen C. and Burns, Mary E., "Trends in Prematurity in the United States", American Journal of Public Health, Vol. 60, 1967 (October 1970).
20. Sternglass, E.J., "Infant Mortality and Nuclear Tests", Bulletin of the Atomic Scientists, Vol. 25, p. 18, April 1969.
21. Sternglass, E.J., "Environmental Radiation and Human Health", in "Effects of Pollution on Health", Vol. 6, in Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability, L.M. LeCam, J. Neyman, and E.L. Scott, eds., pp. 145-221, University of California Press, Berkeley, CA, 1972; see also Sternglass, E.J., "Radioactivity", Chapter XV, Environmental Chemistry, J. O'M. Bockris, ed., Plenum Press, New York, 1977.
22. Monthly Vital Statistics Reports, U.S. Department of Health, Education and Welfare, Vol. 28, No. 1, May 11, 1979.
23. Moriyama, I.M., "Recent Change in Infant Mortality Trend", Public Health Reports 75, 391 (1960); also "Infant Mortality Trends - U.S. and Each State 1930-1964"; National Center for Health Statistics, Ser. 20, No. 1, November 1965.
24. Lindop, J. and Rotblat, A., "Strontium-90 and Infant Mortality", New Scientist, Vol 224, p. 1257, 1970.
25. Glasstone, S., ed., "The Effects of Nuclear Weapons", U.S. Atomic Energy Commission, U.S. Government Printing Office, Washington, DC, 1962.
26. Hull, A.P., "Background Radiation Levels at Brookhaven National Laboratory", Report submitted May 15, 1970, at the Licensing Hearings, Shoreham Nuclear Plant (AEC Docket No. 50-322).

27. Brodsky, A., "Criteria for Acute Exposure to Mixed Fission Product Aerosols", Health Physics, Vol. 11, p. 1017, 1965.
28. Bertel, R., "Maternal-Child Health Effects from Radioactive Particles in Milk, Wisconsin 1963-75", American Public Health Association Annual Meeting, 1979, New York City.
29. Monthly Vital Statistics Reports, National Center for Health Statistics, U.S. Department of Health, Education and Welfare, 3700 East-West Highway, Hyattsville, MD 20782.
30. See Appendix for original data on all states. The District of Columbia has a majority of non-white births and has for decades had an abnormally large infant mortality rate far above the rest of the United States.
31. The Harrisburg Hospital accounts for approximately 30% of all births in the Harrisburg area. The numbers of deaths refer only to those of newly born infants that die within a few hours after birth, and which account for 70-80% of all infant deaths within the first year. Magee Hospital accounts for approximately half of all births in the Pittsburgh area.
32. Sternglass, E.J., "Evidence for Low-Level Radiation Effects on the Human Embryo and Fetus, in Radiation Biology of the Fetal and Juvenile Mammal", Richaldrn, Washington, May 5-8, 1969, M.R. Sikov and D.D. Mahlum, eds., ERDA Symposium Series, Conf-69051, pp. 693-718, 1969.
33. Sternglass, E.J., "Epidemiological Studies of Fallout and Patterns of Cancer Mortality", Hanford Biology Symposium Proceedings, AEC Symposium Series, Vol. 29, June 1973 (Conf-720505).
34. Lave, L.B., Leinhard, S., Kaye, M.B., "Low Level Radiation and U.S. Mortality", Working Paper No. 19-70-1, Graduate School of Ind. Admin., Carnegie-Mellon University, Pittsburgh, PA 15213, July 1971.
35. Sternglass, E.J., and Bell, S., "Nuclear Fallout and the Decline in Aptitude Scores", Presented at the 87th Annual Convention of the American Psychological Association, September 1-5, 1979, New York City.
36. "Radioactive Materials Released from Nuclear Power Plants (1976)", T.R. Decker, U.S. Nuclear Regulatory Commission, Washington, DC 20555, (NUREG-0367) (March 1978), Available from National Tech. Inf. Service, Springfield, VA 22161, Tables 3&5.
37. Beya, J., "Some Long Term Consequences of Hypothetical Major Releases of Radioactivity to the Atmosphere from Three Mile Island", Report to President's Council on Environmental Quality, September 7, 1979, (Center for Energy and Environmental Studies, Princeton University, Princeton, NJ 08544).
38. The maximum permissible discharges can be calculated from information given in earlier editions of Reference 36.

39. Petkav, A., and Chelak, W.S., "Effect of Na-22 on a Phospholipid Membrane", Health Physics, Vol. 22, March 1972.
40. Baum, J.W., "Population Heterogeneity Hypothesis of Radiation Induced Cancer", Health Physics, Vol. 25, p. 97, 1973.
41. The great difference in sensitivity of different population groups is discussed in detail in Reference 14.
42. Morgan, K.Z., "The Non-Threshold Dose-Effect Relationship", Presented at the Academy Forum of the National Academy of Sciences, Washington, DC, September 27, 1979, (Georgia Inst. of Tech., Atlanta, GA 30322).
43. Sternglass, E.J., "Implications of Dose-Rate Dependent Cell-Membrane Damage for the Biological Effect of Medical and Environmental Radiation", in "Population Exposures", Proceedings of 8th Midyear Topical Symposium, Health Physics Society, October 21, 1974 (CONF-741018).
44. E.J. Sternglass, "Cancer Mortality Changes Around Nuclear Facilities in Connecticut" in "Radiation Standards and Public Health", Proceedings of a Second Congressional Seminar on Low-Level Ionizing Radiation, February 10, 1978 (Environmental Policy Institute, 317 Pennsylvania Avenue, S.E., Washington, DC 20003).
45. "Annual Environmental Reports to the N.R.C.", Millstone Point Nuclear Power Station, Unit I (North-East Utility Company). Available at the Regional Office, N.R.C., King of Prussia, Pennsylvania.
46. Letter to Congressman Christopher J. Dodd from Harold P. Denton, Director, Office of Nuclear Reactor Regulation, N.R.C., July 27, 1978.
47. Letter to Congressman Christopher J. Dodd from Douglas M. Costle, Administrator of the Environmental Protection Agency, August 9, 1978.
48. Sternglass, E.J., "A Preliminary Report on Changes in Infant Mortality Patterns Following the Arrival of Fallout from the September 26, 1976 Chinese Nuclear Weapons Test", Presented at a meeting of the National Academy of Science Committee on the Effect of Ionizing Radiation, Washington, DC, July 18, 1977.

TABLE I

Infant Mortality Changes in Pennsylvania Following the
Three Mile Island Accident on March 28, 1979

<u>1979</u>	<u>No. of Infant Deaths</u>	<u>No. of Live Births</u>	<u>Rate 1000</u>	<u>Average Rate</u>	<u>% Change in Rate</u>
February	147	11,892	12.4		
March	141	13,589	10.4	11.9	
April	166	12,520	13.3		
May	198	13,201	15.0		
June	163	12,293	13.3	15.7	+32%
July	271	14,680	18.5		

Source: U.S. Monthly Vital Statistics

TABLE II

Infant Mortality East of the Mississippi in July of 1979.

1979

	Infant deaths	Live births	Rate per 1000
Pennsylvania	271	14,680	18.5
New York ⁺	295	22,855	12.9
N.J.	83	8,041	10.3
Maine	12	1,440	8.3
N. Hampshire	8	1,089	7.3
Vermont	5	513	9.7
Mass.	54	6,690	8.1
Rhode Island	16	1,149	13.9
Conn.	15	3,681	4.1
Ohio	147	14,071	10.4
Indiana	101	7,885	12.8
Illinois	N.A.	N.A.	N.A.
Michigan	147	12,346	11.9
Wisconsin	48	6,272	7.7
Delaware	10	830	12.0
Md.	66	4,179	15.8
D.C.*	22	1,133	19.6*
Virginia	91	6,535	13.9
W. Va.	27	2,520	10.7
N. Carolina	104	7,660	13.6
S. Carolina	59	4,253	13.9
Georgia	89	6,700	13.3
Florida	164	10,362	15.8
Kentucky	50	5,616	8.9
Tennessee	81	6,281	12.9
Alabama	60	4,893	12.3
Mississippi	59	4,253	13.9
New York City	150	9,300	16.1
U.S.	3,800	306,000	12.4

+ Includes N.Y. City

* Majority of population is non-white and has had much above average infant mortality for decades.

Table III

Comparison of Changes in Infant Mortality Rates Following
the Three Mile Island Accident in Harrisburg and Adjoining Areas
at Increasing Distance East and West of Harrisburg

	Feb., March, April		May, June July		% Change in Death Rate	Approximate Distance in Miles
	<u>No.</u>	<u>Rate</u>	<u>No.</u>	<u>Rate</u>		
Ohio	482	12.0	480	11.7	-3%	200m West
Pennsylvania	454	11.9	632	15.7	+32%	180m West
Pittsburgh ^f	31	14.2	52	23.4	+65%	180m West
Harrisburg ⁺	1	1.9	7	13.9	+630%	10-20m North
Upstate New York	404	13.0	477	14.3	+8%	100m Northeast
New York City	429	17.1	438	16.4	-4%	200m East
United States		14.0		12.6	-10%	200m

PA and NY State 1769 2027

Change in No. + 258

^f For Magee Hospital with 50% of births in the area

⁺ For Harrisburg Hospital with 35% of births in the area

TABLE IV

Newborn Mortality Rate Changes in the Harrisburg Hospital in 3 Months Period Before and After the Three Mile Island Accident March 28, 1979*, Compared with the Same Periods in 1978.

1978	No. of Infant Deaths	No. of Live Births	Rate 1000	Average Rate	%Change in Rate
February	0	128	0.0		
March	1	187	5.3	2.2	
April	0	137	0.0		
May	1	175	5.7		
June	0	154	0.0	2.1	-5%
July	0	154	0.0		
<u>1979</u>					
February	1	186	5.4		
March	0	172	0.0	1.9	
April	0	167	0.0		
May	2	166	12.0		
June	4	154	26.0	13.9	+630%
July	1	182	5.5		

* This hospital accounts for about one-third of all deliveries in the Harrisburg area. Newborn infant mortality is defined as a live-born infant that dies within a short time, most die within a few hours. (Data from monthly Pediatric Conference Records).

Table V

Newborn Infant Mortality Rate Changes in the Pittsburgh Area
From Hospital Records for Three Months Before and After the Three Mile
Island Accident March 28, 1979, Compared With the Same Period in 1978

	<u>No. of Infant Deaths</u>	<u>No. of Live Births</u>	<u>Rate 1000</u>	<u>Average Rate</u>	<u>% Change in Rate</u>
<u>1978</u>					
February	7	633	11.0		
March	6	750	8.0	14.4	--
April	16	625	25.6		
May	11	719	15.3		
June	8	684	11.7	12.5	-13%
July	8	763	10.5		
<u>1979</u>					
February	10	682	14.7		
March	11	786	14.0	14.2	--
April	10	721	13.9		
May	21	664	31.6		
June	7	759	9.2	23.4	+65%
July	24	798	30.1		

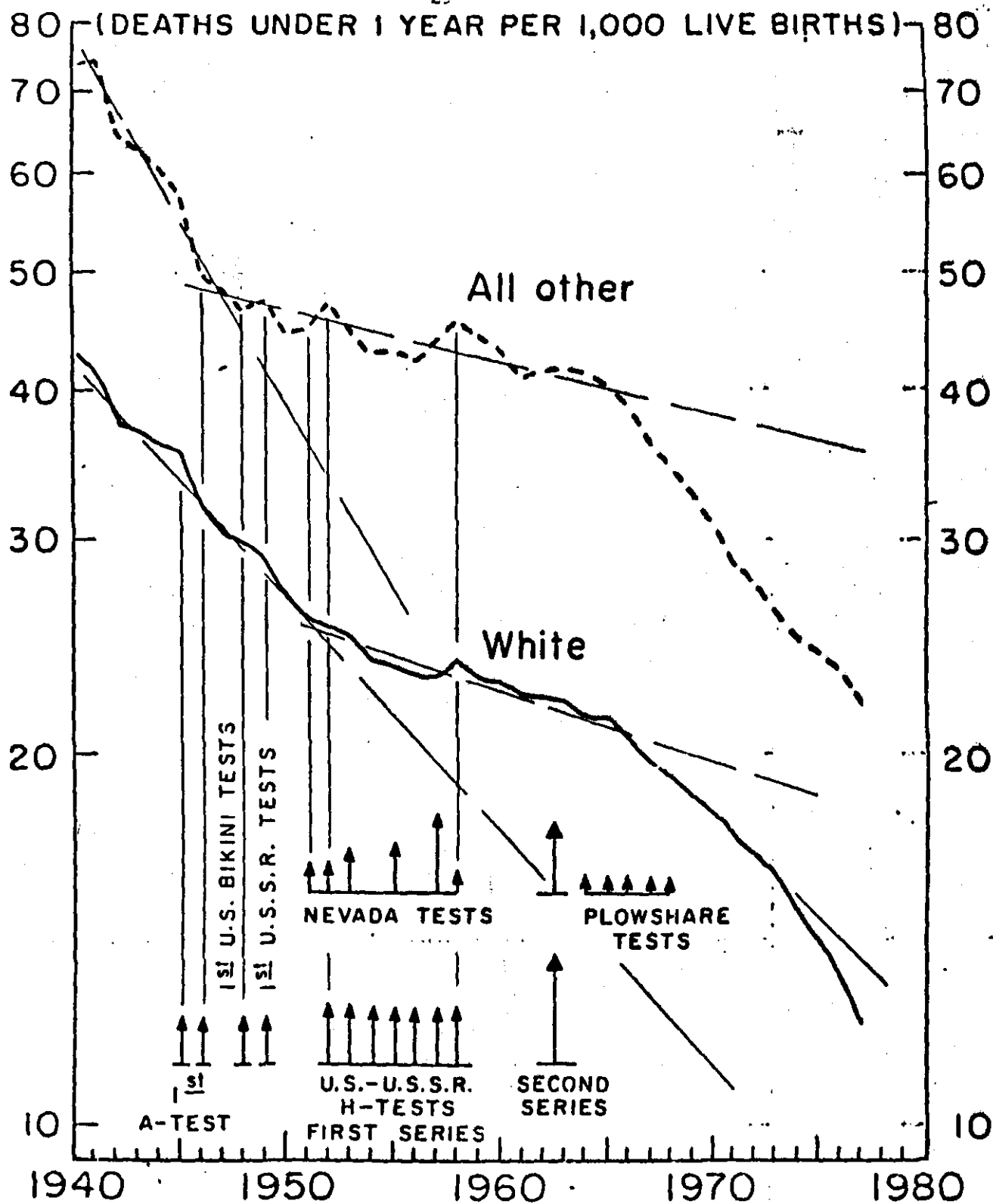
+ Data are from Magee Womens Hospital, which handles approximately one-half of all Pittsburgh area deliveries. Newborn infant death is defined as a live-born infant that dies within one month; most die within hours after birth. (Data supplied by Department of Pathology.)

Table VI

Infant Mortality Changes In Northern and Southern New England in
July 1979 After Three Mile Island Accident Compared with July 1978

	JULY 1978			JULY 1979			% Change in Rate
	<u>No. of Deaths</u>	<u>No. of Live Births</u>	<u>Rate per 1000</u>	<u>No. of Deaths</u>	<u>No. of Live Births</u>	<u>Rate</u>	
<u>Northern New England States</u>							
New Hampshire	2	1036	1.9	8	1089	7.3	+284%
Vermont	3	711	4.2	5	513	9.7	+131%
Maine	10	1306	7.7	12	1440	8.3	+8%
Northern New England	15	3053	4.9	25	3042	8.2	+67%
<u>Southern New England States</u>							
Connecticut	27	3179	8.5	15	3681	4.1	-52%
Rhode Island	21	1054	19.9	16	1149	13.9	-30%
Massachusetts	44	5556	7.9	54	6690	8.1	+3%
Southern New England	92	9789	9.4	85	1152	7.4	-21%

Source: U.S. Monthly Vital Statistics



FROM: U.S. MONTHLY VITAL STATISTICS
REPORTS; VOL. 28, NO. 1
MAY 11, 1979

Figure 1. Infant Mortality Rates By Color: United States, 1940-1977

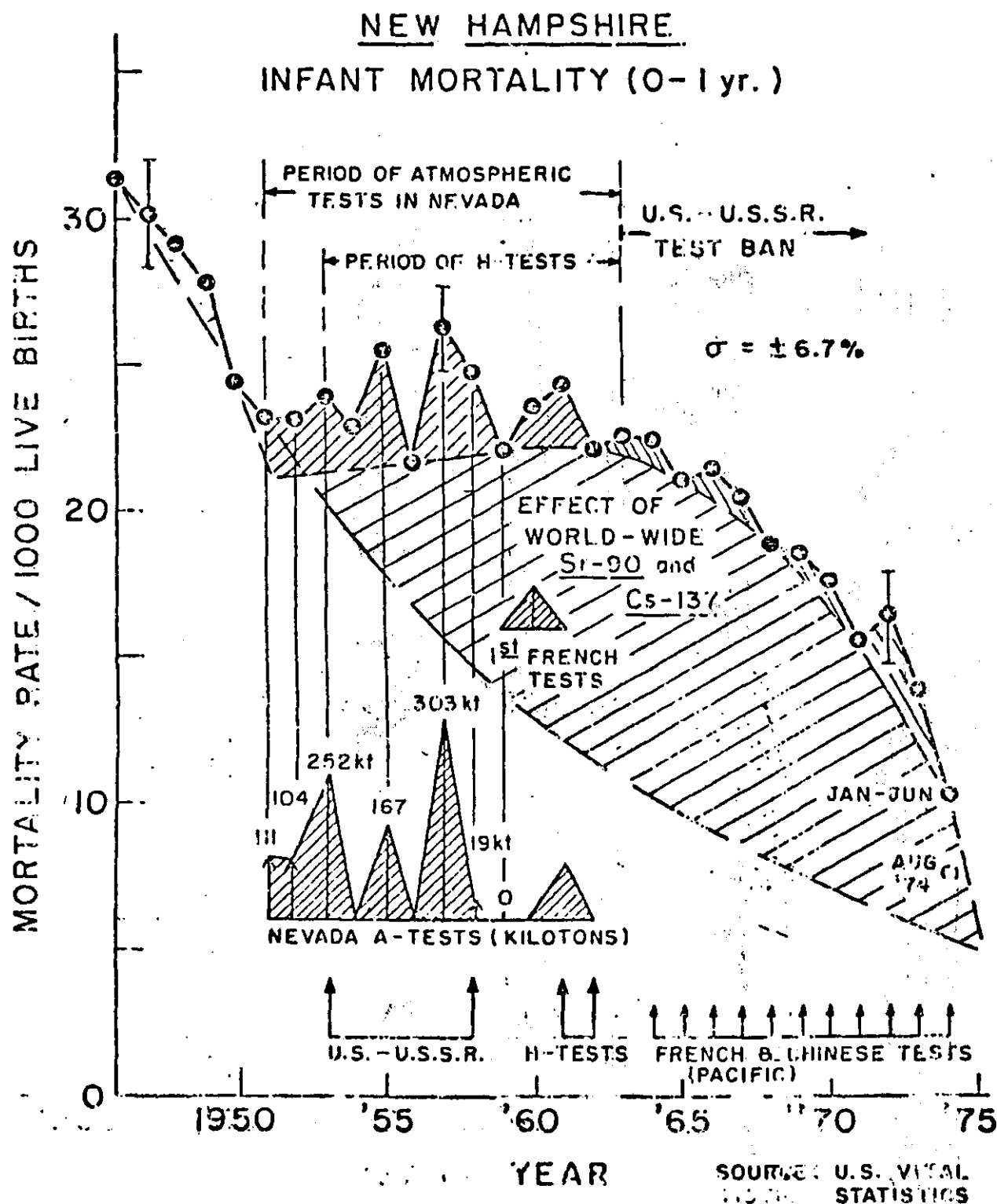


Figure 2. Infant mortality rate in New Hampshire between 1945 and 1974 together with announced yields of atmospheric nuclear weapons tests in Nevada. Note close correspondence between the sharp peaks in mortality rates and the peaks of weapons test yields in Nevada.

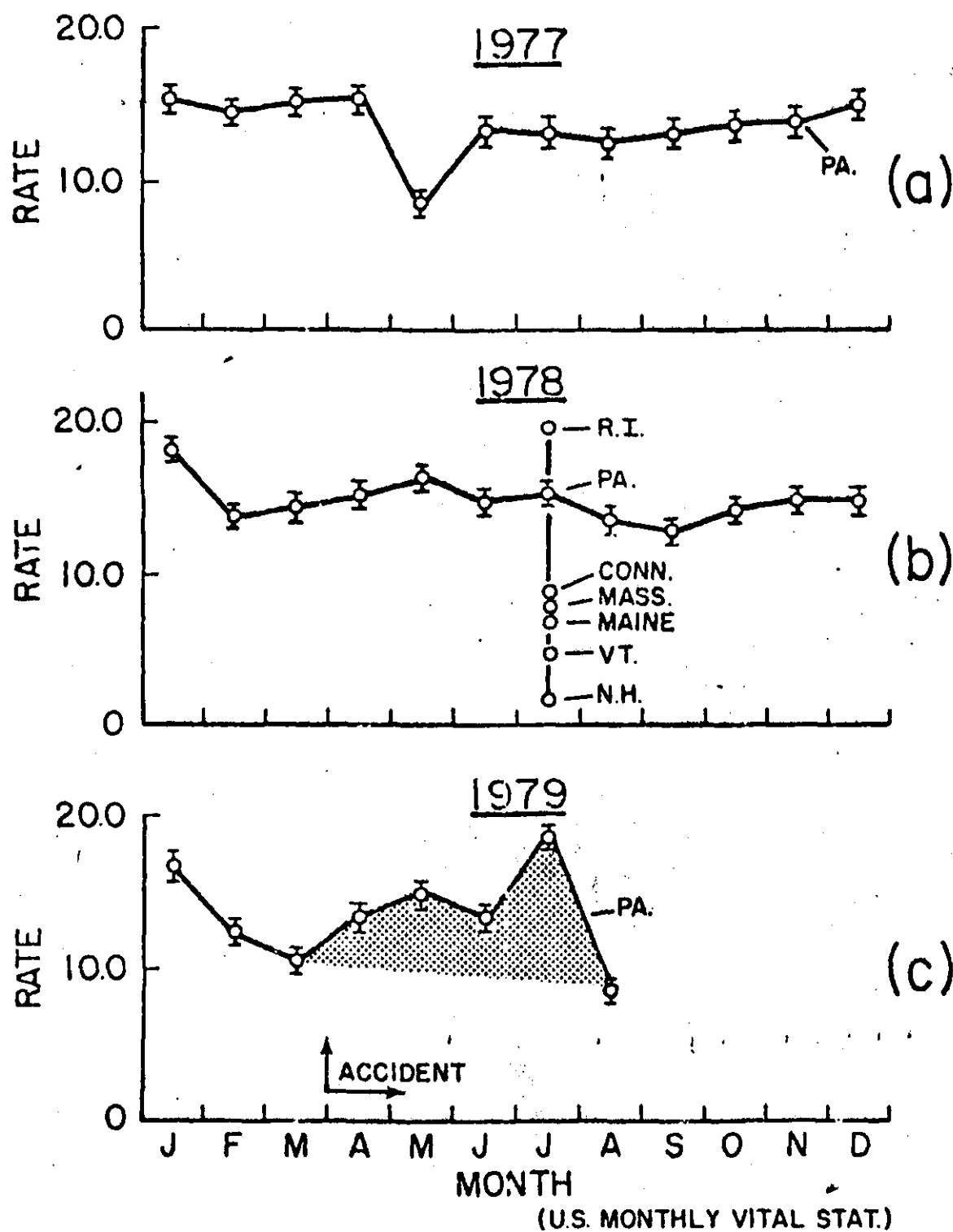


Figure 3. Monthly infant mortality rates in Pennsylvania before and after the Three Mile Island accident of March 28, 1979. (a) 1977, (b) 1978 and (c) 1979. In (b) the July 1978 infant mortality rates have been plotted for the New England states. Note that Rhode Island downwind from the Millstone Reactor had the highest mortality rate, and distant New Hampshire the lowest.

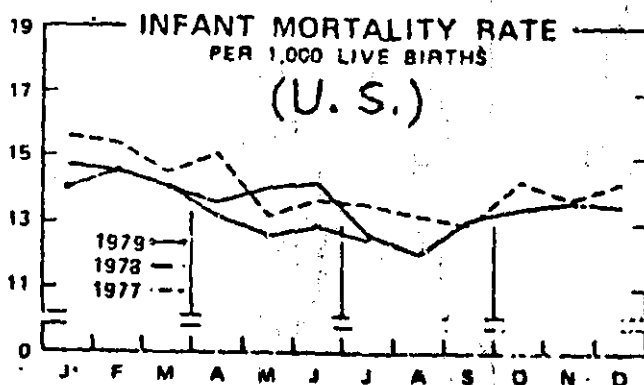


Figure 4. Monthly U.S. infant mortality rate for 1977, 1978 and 1979 (to July).
Note normal pattern of low infant mortality rates in the summer months.

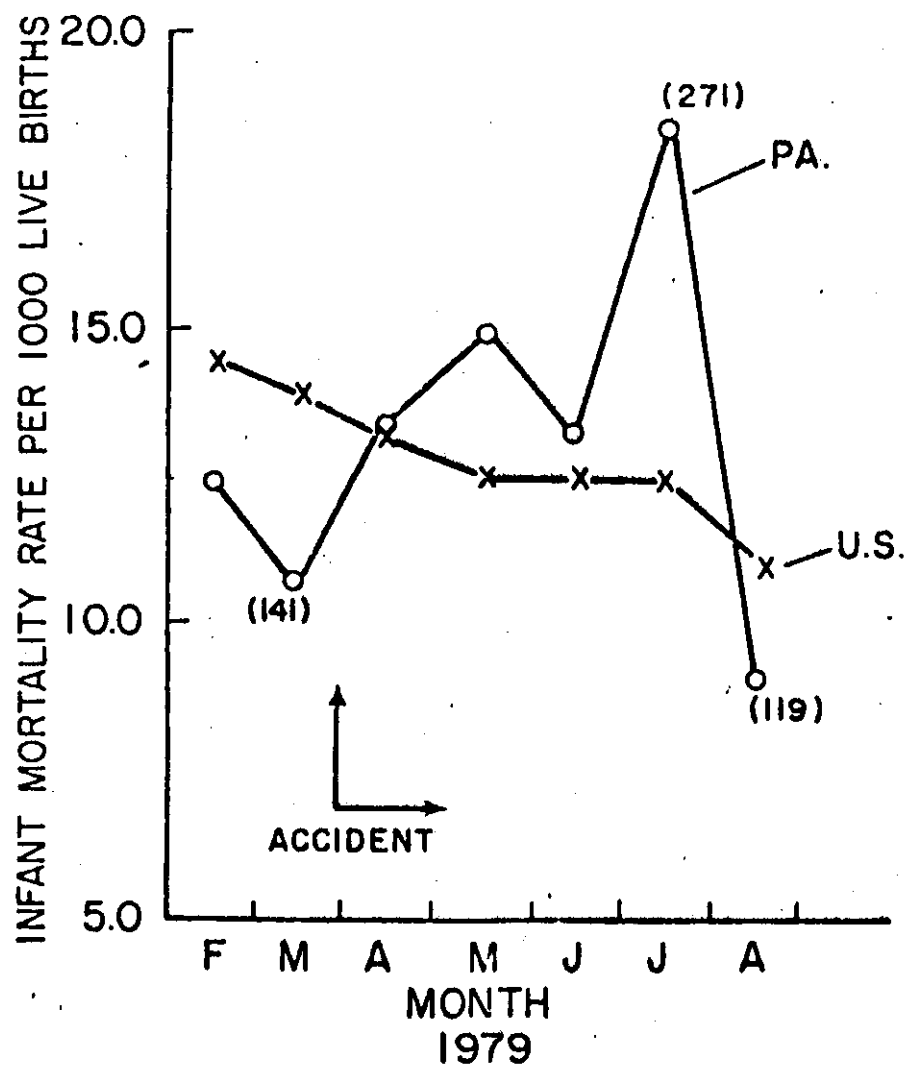


Figure 5. Comparison of the monthly infant mortality rates for Pennsylvania and the United States as a whole for February to August 1979, before and after the Three Mile Island accident on March 28. Figures in brackets indicate number of infant deaths.

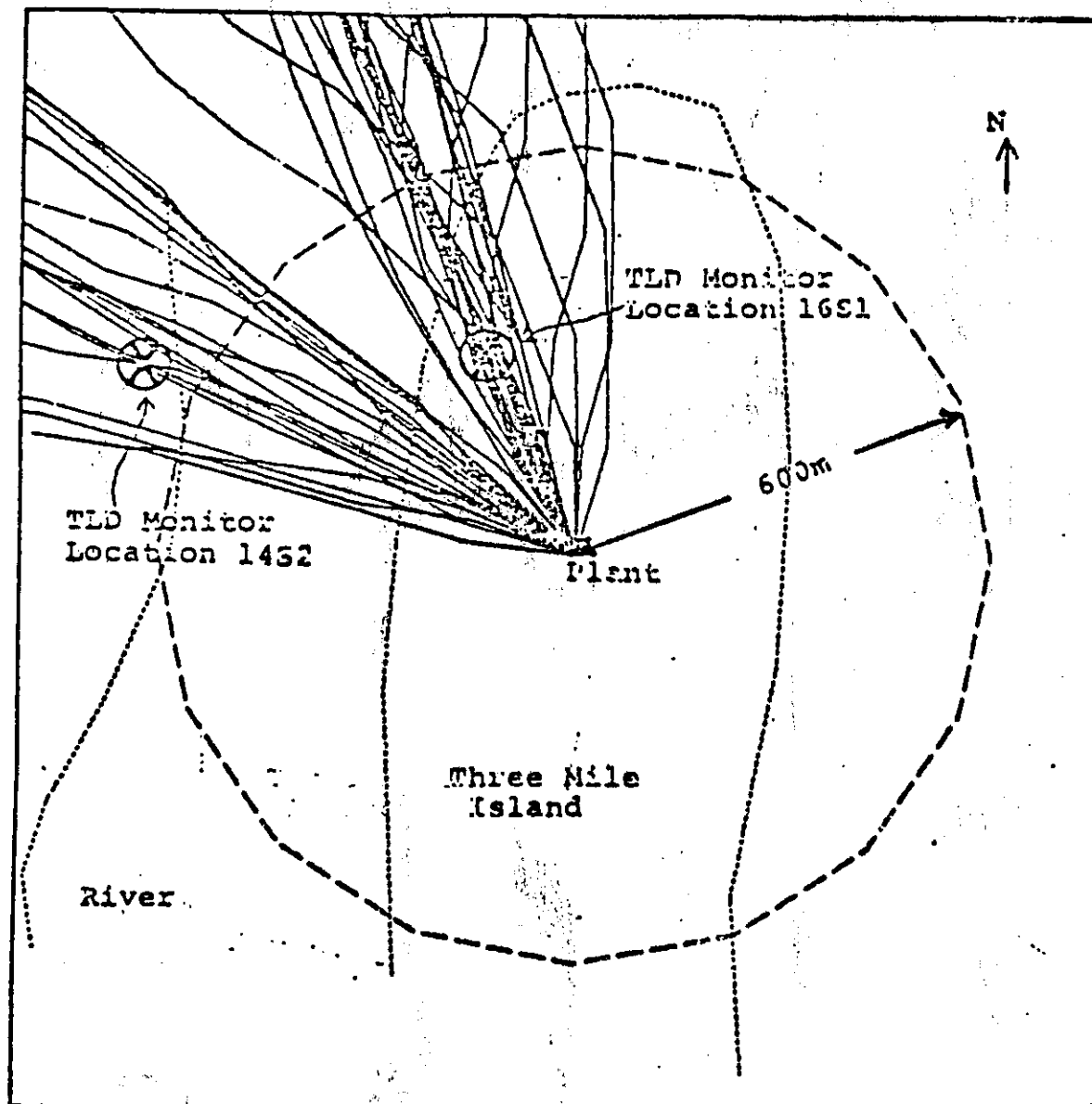


Figure 6. Approximate tracks of plume centerlines for the radioactive gases released during the Three Mile Island accident every 15 minutes starting at 5 p.m. March 29, 1979. (Based on TMI Outsite, Meteorological Tower Data, Reference 4, Figures 4-5).

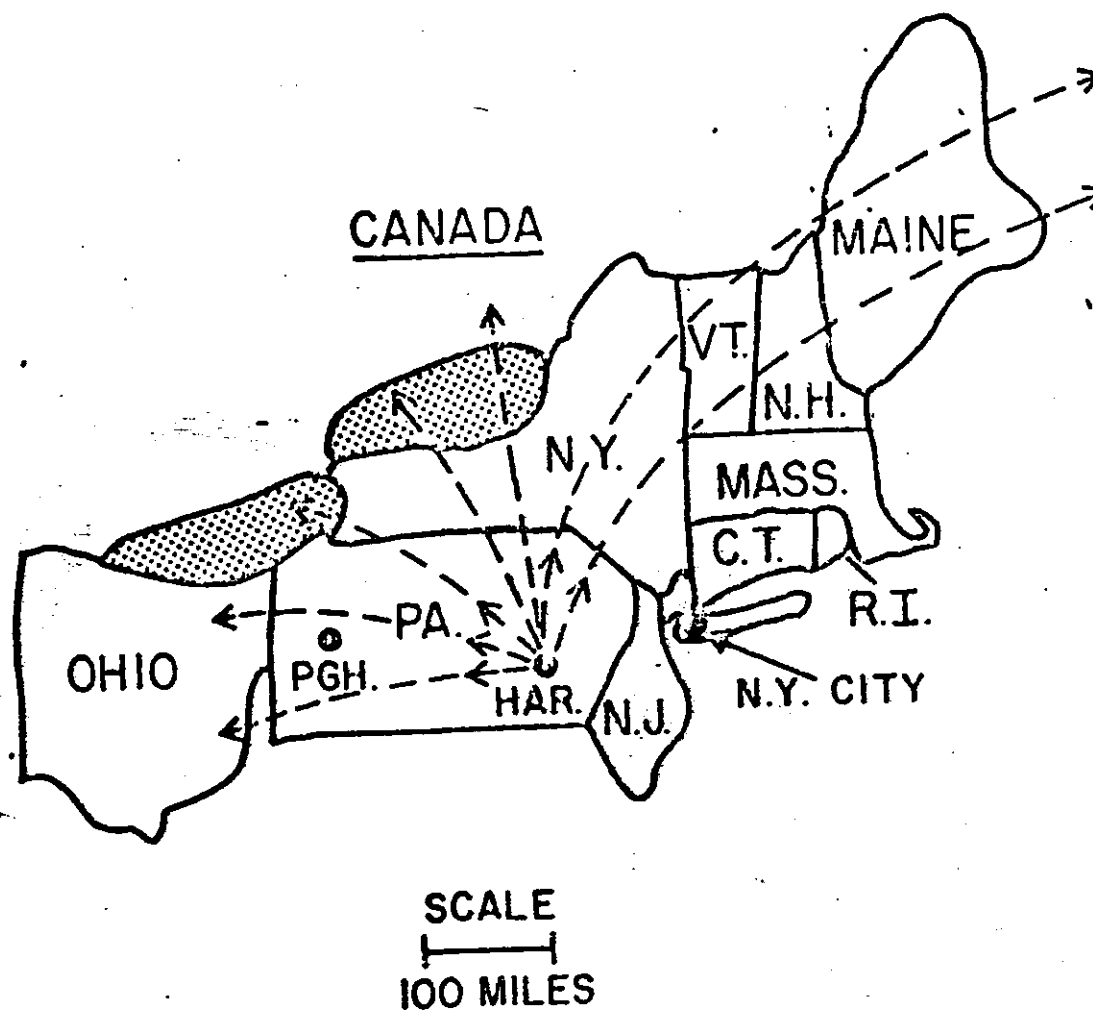


Figure 7. Map of the north-eastern United States showing location of Harrisburg, Pittsburgh and neighboring states together with the principal directions in which the radioactive gases drifted in the first two days of the accident at Three Mile Island.

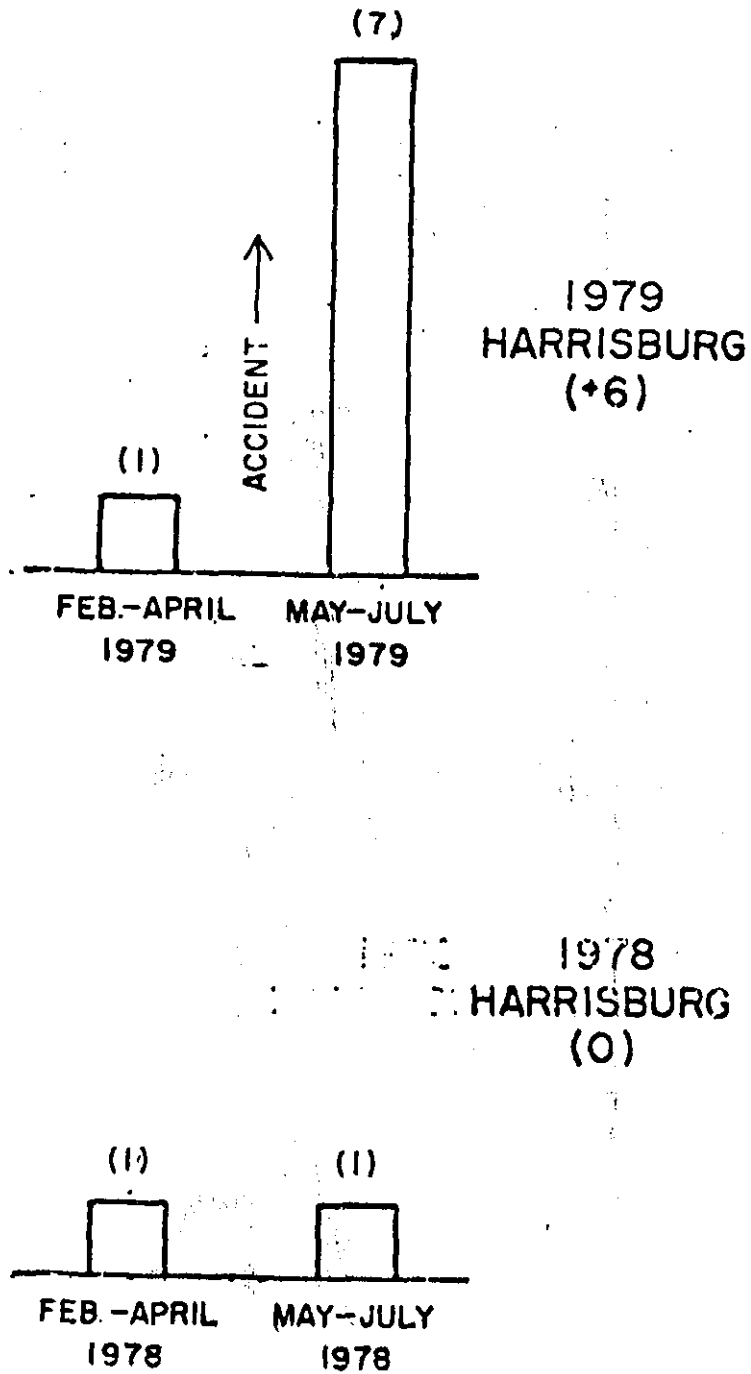


Figure 8. Newborn infant deaths in the Harrisburg Hospital for three month periods before and after the accident at Three Mile Island, March 28, 1979. Figures in brackets represent the change in the number of deaths. Note the absence of any increase in May/June/July of 1978.

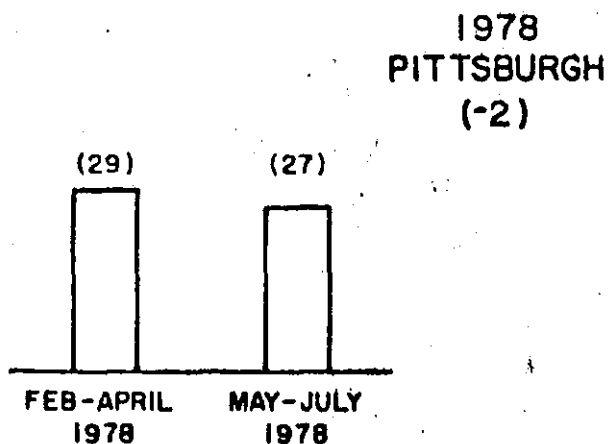
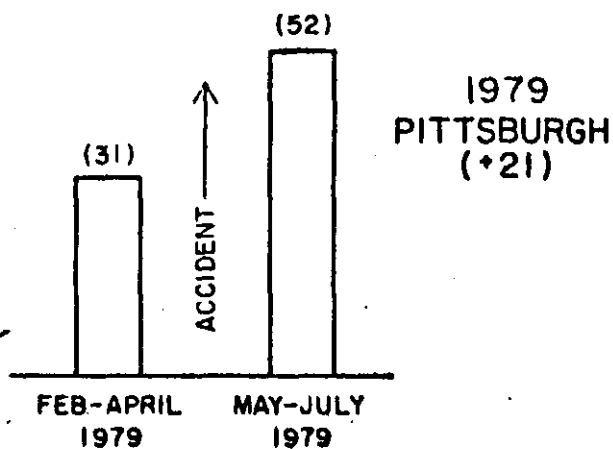


Figure 9. Newborn infant deaths in Pittsburgh (Magee Hospital) for 3 month periods before and after the Three Mile Island Accident of March 28, 1979. Note the absence of any rise in May/June/July of 1978.

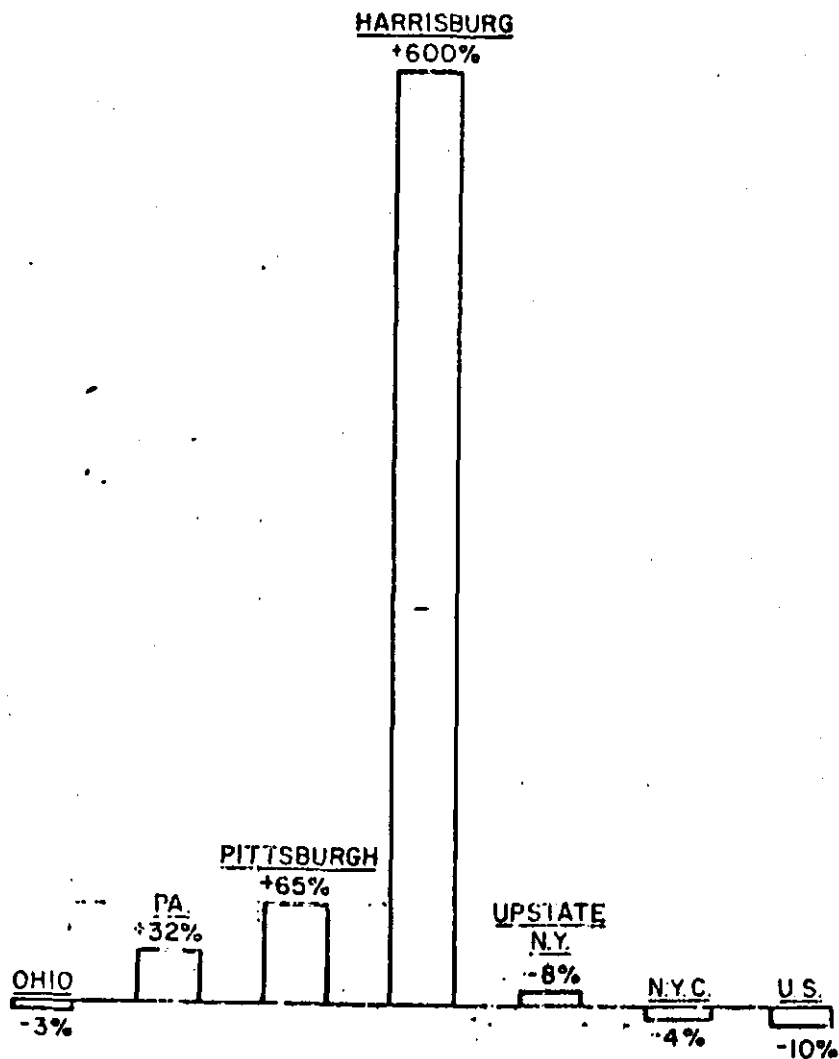


Figure 10. Geographical pattern of newborn infant mortality changes at various distances from the Three Mile Island plant located 10 miles south of Harrisburg. Figures represent percent changes between February/March/April and May/June/July 1979. Note that greatest rises occurred closest to the plant.

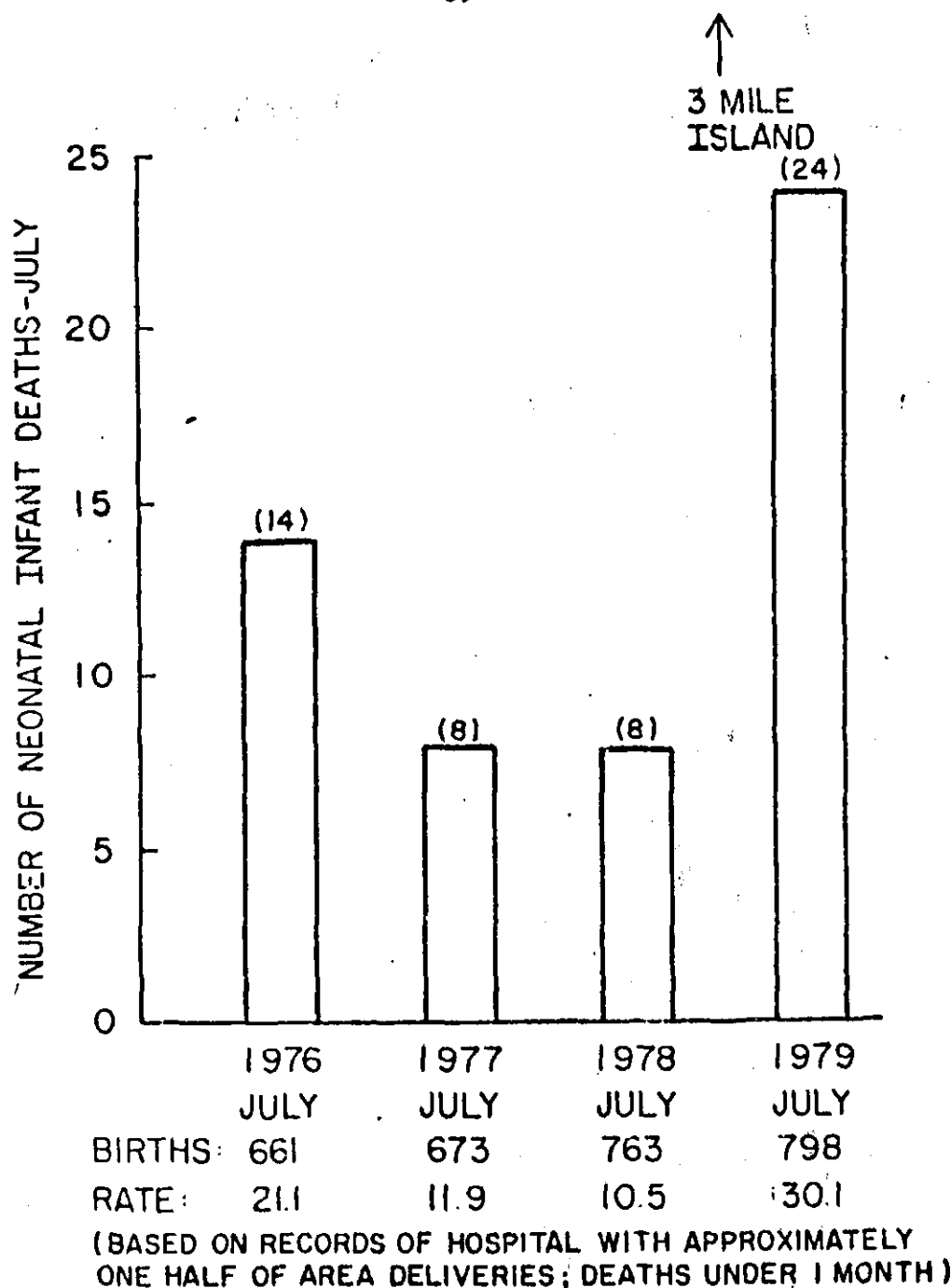


Figure 11. Trend in the July infant mortality rate in the Pittsburgh area before and after the Three Mile Island accident in March 1979. Numbers in brackets refer to the number of newborn infant deaths at Magee Hospital for the month of July 1976, 1977, 1978 and 1979. Note that the death-rate tripled between July 1978 and July 1979.

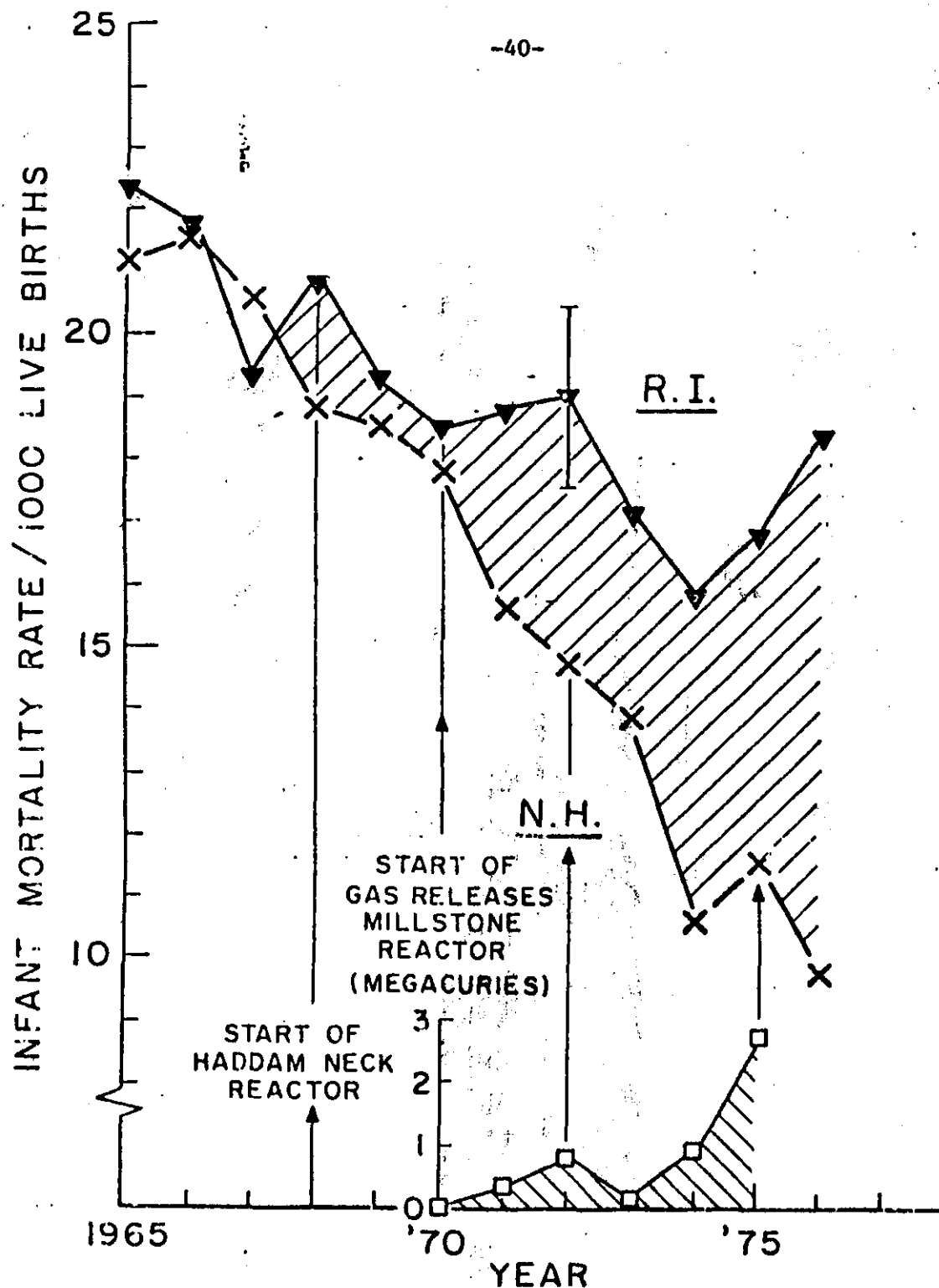


Figure 12. Infant mortality rate trends in Rhode Island 20-40 miles downwind from the Millstone Reactor in Connecticut between 1965 and 1975 compared with the rate for New Hampshire more than 100 miles to the north. Also shown are the annual releases of gaseous radioactivity in millions of curies. Note the widening gap between the two states indicated by the shaded area following the start of nuclear reactor operations.

ΔR -PERCENT EXCESS INFANT MORTALITY - R.I. vs N.H.

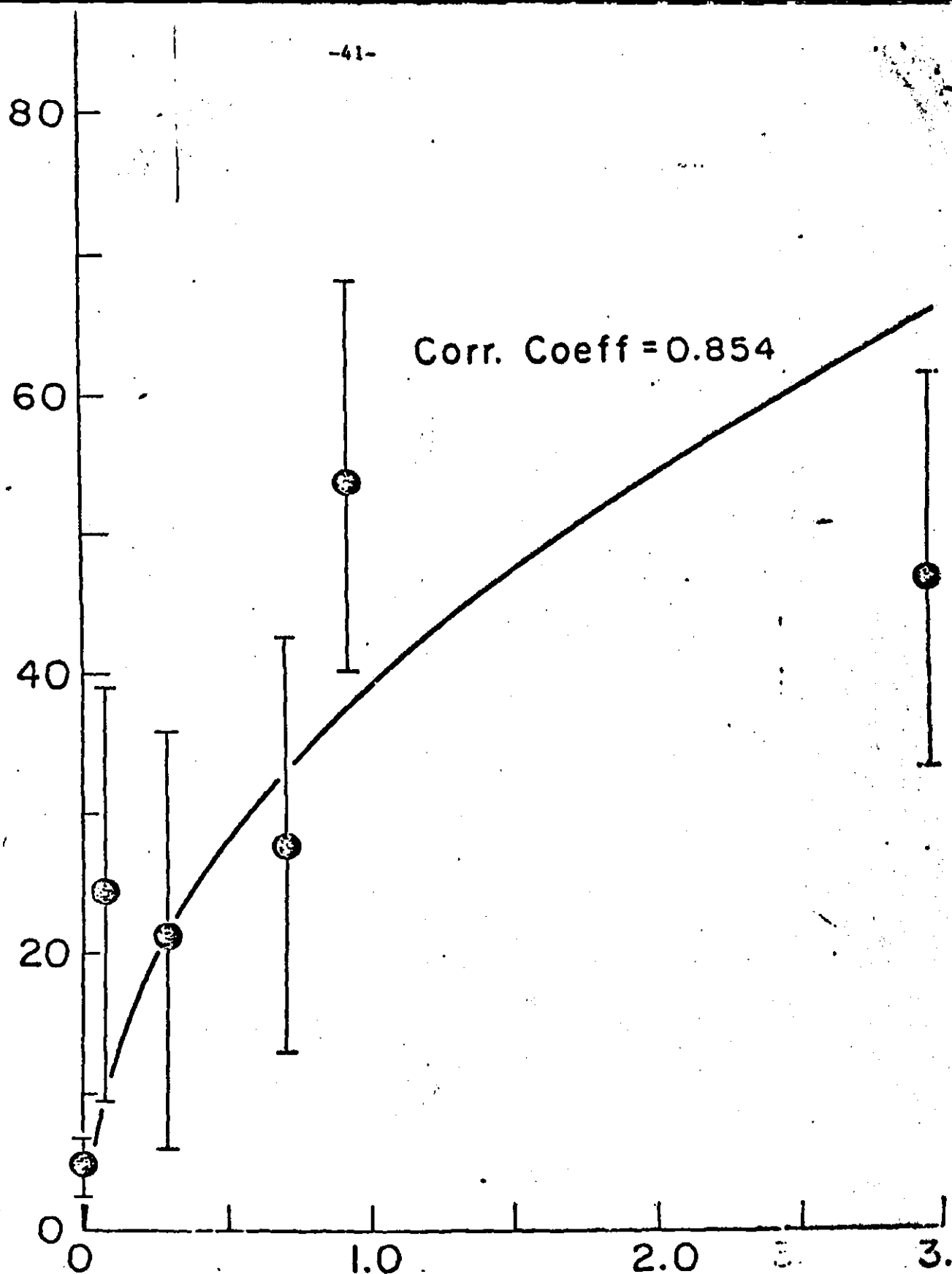


Figure 13. Percent change in the excess infant mortality in Rhode Island relative to the more distant state of New Hampshire as a function of the total gaseous radioactivity releases from the Millstone plant near the Rhode Island border taken from Figure 12. (Two year moving averages). Note that the data points are best fitted by a curve that rises more rapidly at low doses than at high doses.

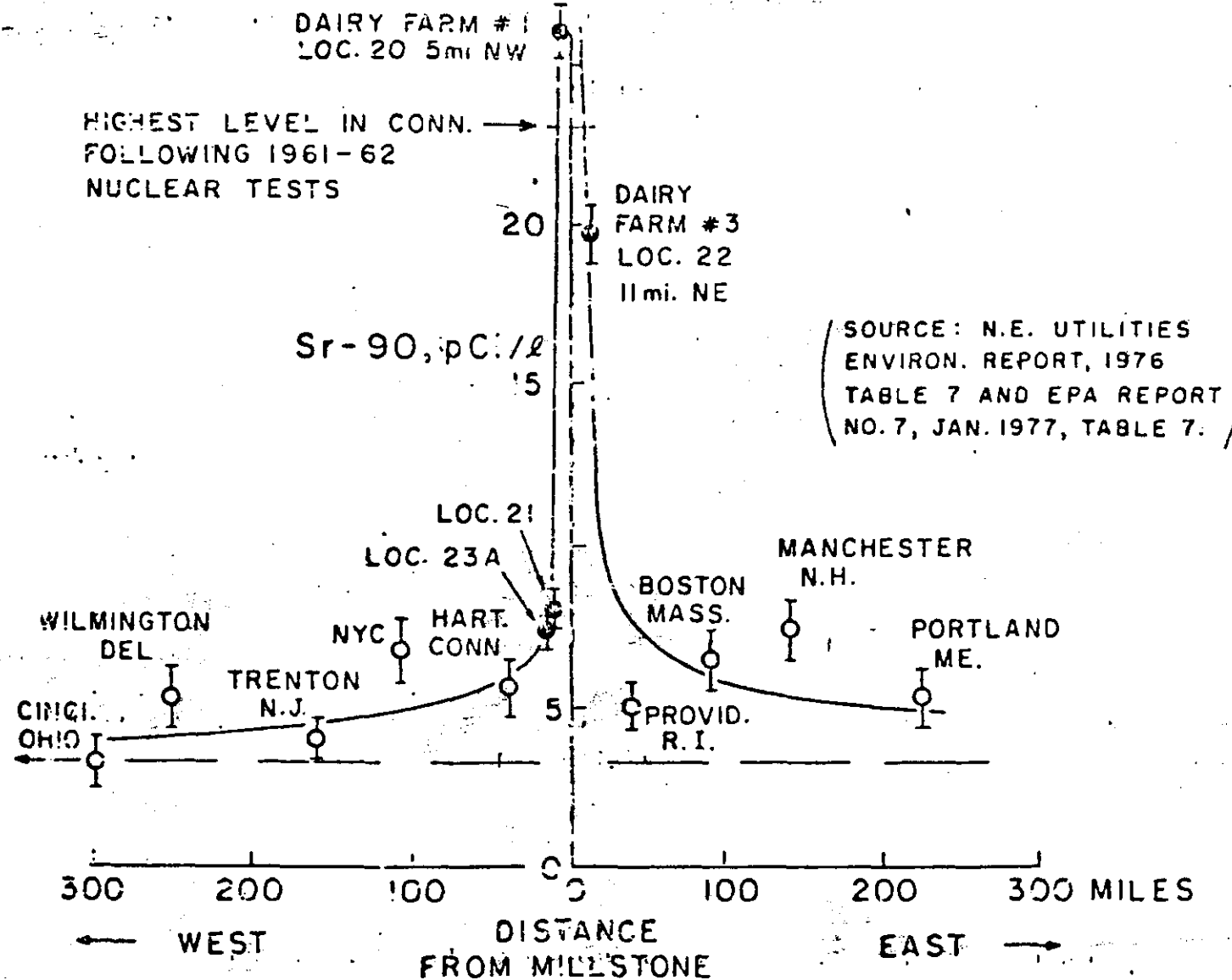


Figure 14. Geographical pattern of Strontium-90 in milk at different distances from the Millstone Nuclear plant near Waterford, Connecticut for July 1976. Solid points represent the samples collected by the utility, open points represent EPA measurements. Note that the concentrations measured nearest the plant are close to those measured during heavy nuclear weapons testing in the early 1960's.

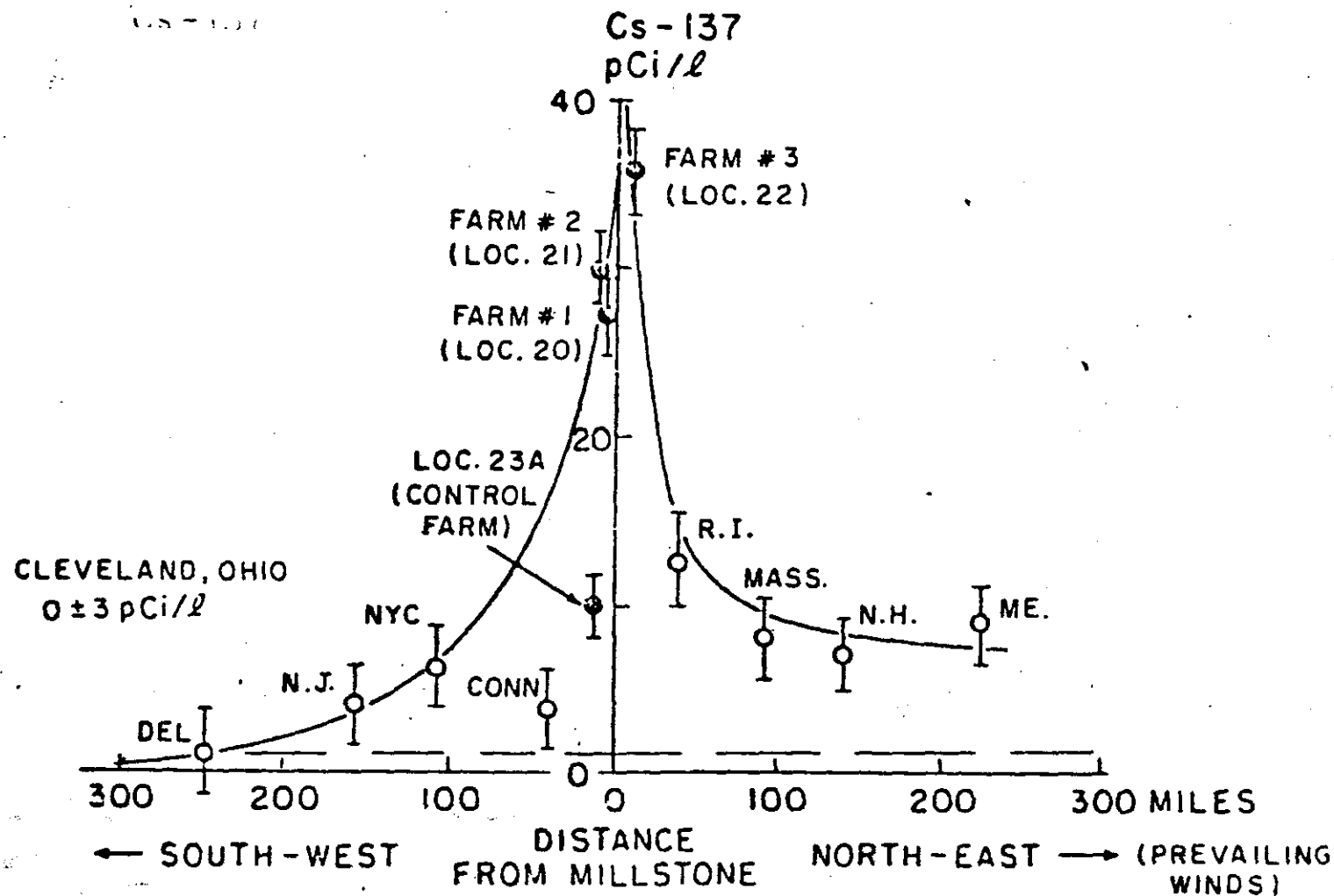


Figure 15. Geographical pattern of Cesium-137 concentrations in the milk at different distances from the Millstone Nuclear plant in August of 1976. Solid points represent data collected by the utility, open points EPA data. Note that highest levels near the plant are three times greater than for the control form (Location 23A), and as much as 10 times greater than for the upwind locations in New Jersey and New York City.

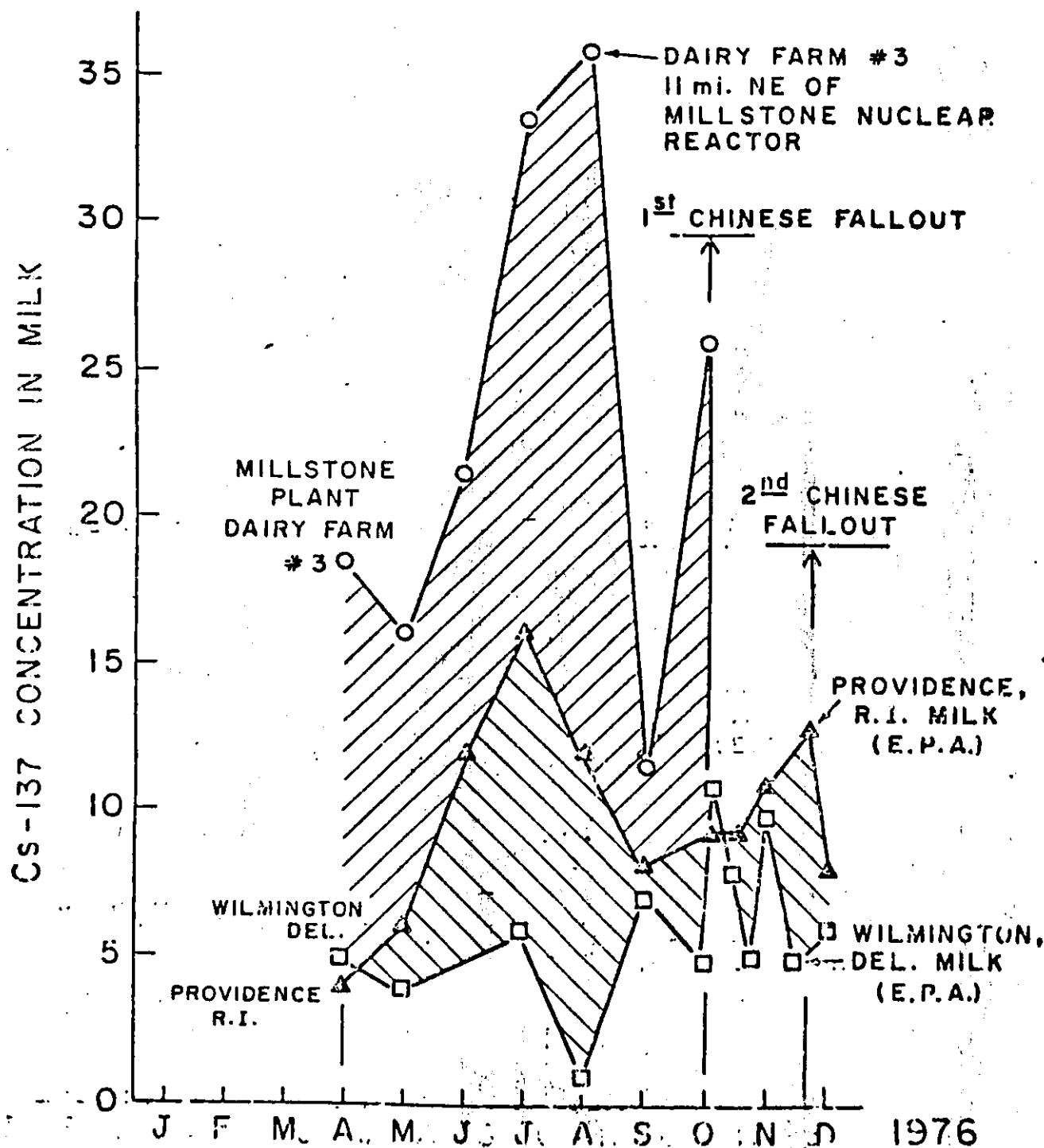


Figure 16. Changes in monthly values of Cesium-137 concentrations in milk in 1976 for three different locations at different distances from the Millstone Nuclear Reactor in Waterford, Connecticut. Note that the largest value occurred closest to the plant in August 1976, the next largest in Providence, Rhode Island, and the lowest in Wilmington, Delaware far to the south. Furthermore, it occurred before the nuclear test by China in September. Also note that the peak associated with the arrival of the heavy fallout from the September 1976 Chinese atmospheric bomb test has a lower magnitude than the July/August maximum measured near the plant, further indicating that the high Cs-137 levels near the plant cannot be attributed to fallout from Chinese bomb tests.

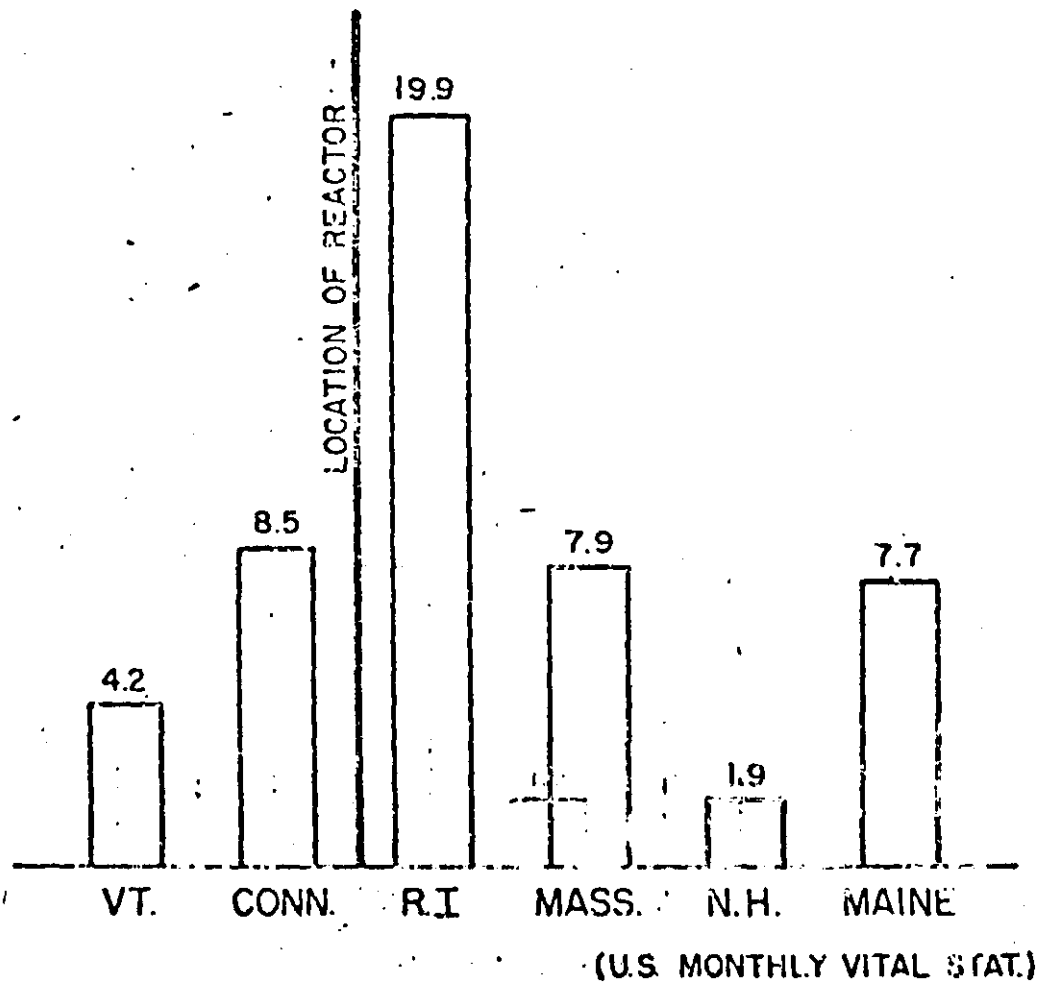
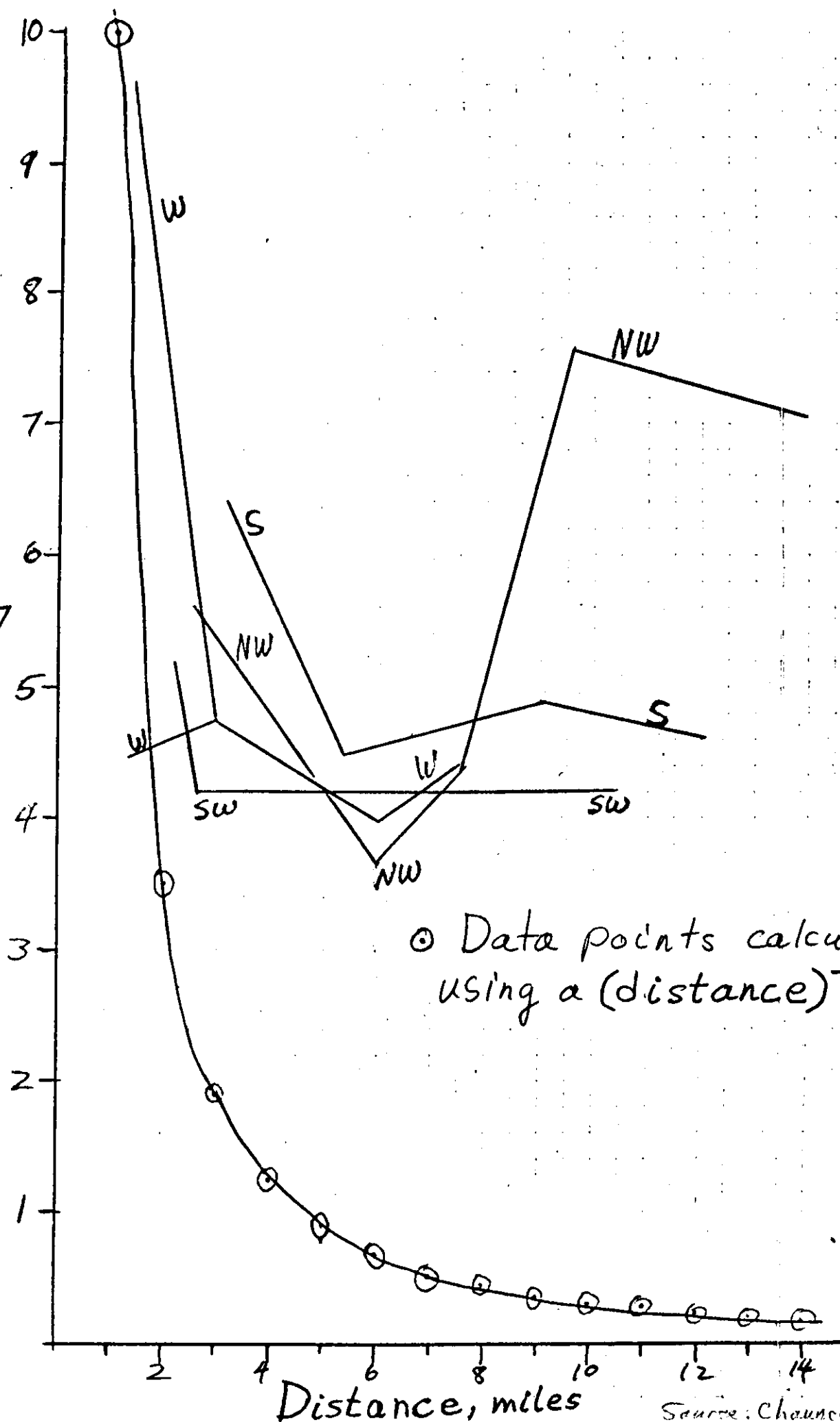


Figure 18. Geographical pattern of infant mortality changes at different distances from the Millstone Nuclear Power Station located near the border between Connecticut and Rhode Island. Note the high rate downwind from the plant in Rhode Island.

Total dose,
mR,
3/31-4/7



⊙ Data points calculated
using $a(\text{distance})^{-1.5}$

Total
dose,
mR,
3/31-4/7

